

Autonomous Data Scientist

Automated Machine Learning, Explainability and Model Evaluation Report

Executive Summary

AutoDS completed automated model development, advanced optimization and final model evaluation.

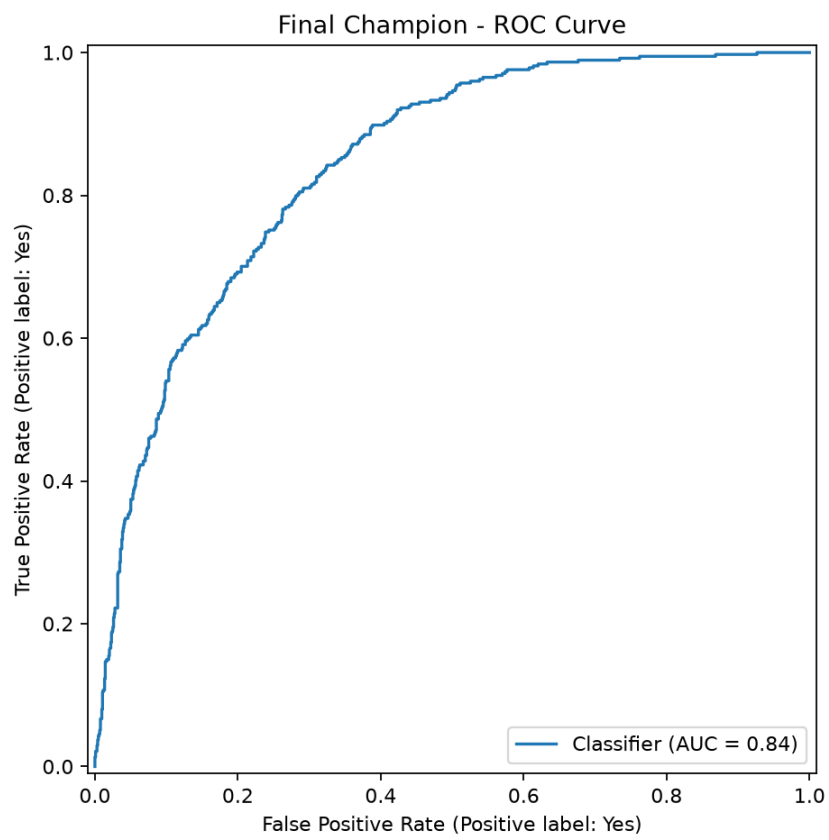
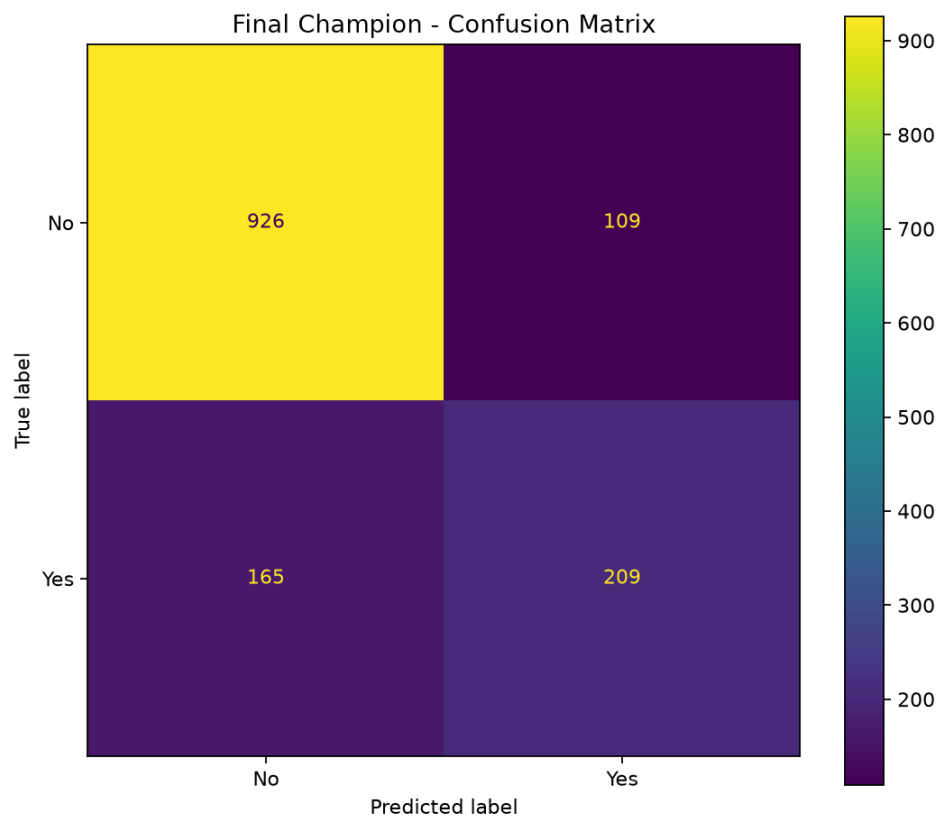
Dataset Overview

Attribute	Value
Rows	7043
Columns	21
Data Quality Score	98.0
Problem Type	classification

Final Model Performance

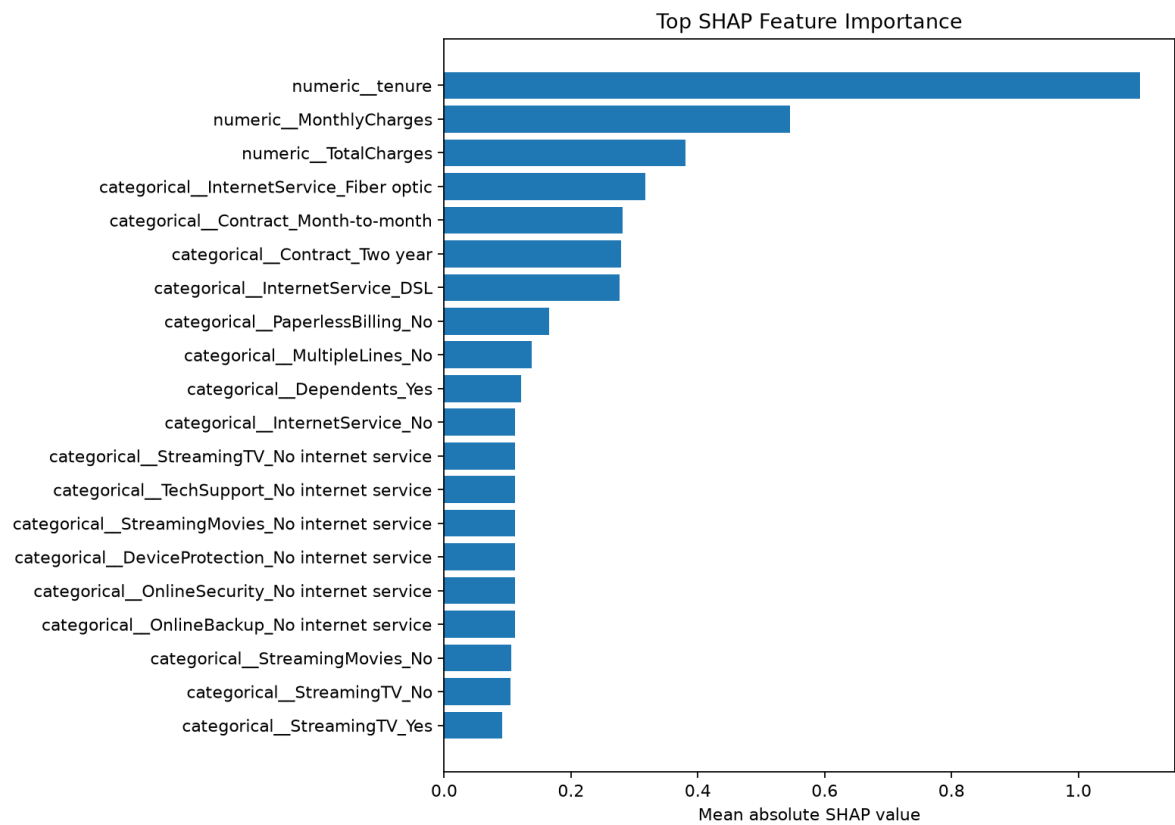
Metric	Value
accuracy	0.8055
precision_macro	0.753
recall_macro	0.7268
f1_macro	0.7376
f1_weighted	0.8002
roc_auc	0.8419

The final verified metrics are available in the model evaluation section.



Model Explainability

Model explanation artifacts were generated using the available feature-importance methods.



Top Model Features

Feature	Importance
numeric__tenure	1.097267
numeric__MonthlyCharges	0.544889
numeric__TotalCharges	0.380365
categorical__InternetService_Fiber optic	0.316941
categorical__Contract_Month-to-month	0.281125
categorical__Contract_Two year	0.278879
categorical__InternetService_DSL	0.277142
categorical__PaperlessBilling_No	0.165314
categorical__MultipleLines_No	0.138304
categorical__Dependents_Yes	0.12139
categorical__InternetService_No	0.111232
categorical__StreamingTV_No internet service	0.111232
categorical__TechSupport_No internet service	0.111232
categorical__StreamingMovies_No internet service	0.111232
categorical__DeviceProtection_No internet service	0.111232

Data Quality Interpretation

Data-quality findings should be reviewed together with the stored quality report.

- Dataset contains missing values.
- Probable identifier columns detected and excluded from training.
- High-cardinality categorical columns detected.

Limitations

- Results depend on the supplied dataset.
- Model relationships should not be interpreted as proof of causation.

Recommendations

- Validate the final model on new unseen data.
- Monitor performance after deployment.